

CKKS HE in a simplified form

Secret key $\rightarrow s = [4, 9] \rightarrow$ in real polynomial
in the form

Modulus $q = 100,003$
scaling factor $\Delta = 1000$

$$R_q = \frac{z_q[x]}{x^N + 1}$$

PT weights

client 1 = 0.10

client 3 = 0.09

client 2 = 0.12

$$b = a \cdot s + e + \Delta \cdot PT \pmod{q}$$

where

$a \rightarrow$ random vec

$e \rightarrow$ random noise.

$$s = [4, 9]$$

C-1 PT = 0.10

$$a_1 = [37, 52]$$

$$e_1 = 5$$

$$b_1 = (37 \times 4 + 52 \times 9) \\ + 5 + 100$$

$$CT = (a_1, b_1)$$

$$CT_1 = ([37, 52], 721)$$

C-2 PT = 0.12

$$a_2 = [21, 66]$$

$$e_2 = 3$$

$$b_2 = 801$$

$$CT_2 = ([21, 66], 801)$$

C-3

$$PT = 0.09$$

$$a_3 = [58, 14]$$

$$e_3 = 4$$

$$b_3 = 452$$

$$CT_3 = ([58, 14], 452)$$

$a \rightarrow$ Public but s is private to clients

Aggregation part on server

$$a_sum = [116, 132] \quad b_sum = 1974$$

Decrypt

$$\hat{i} = a_{\text{sum}} \cdot S = (116 \times 4) + (132 \times 4) = 1652$$

$$r = b - \hat{i} = 322$$

$$\text{res} = r / \Delta = 322 / 1000 = 0.322$$

$$\text{Divide by no of clients} = 0.322 / 3 = 0.10733$$

$$\text{updated weight} = 0.10733$$

Compare with real calculation w/o this enc

$$\text{True Avg} = (0.10 + 0.12 + 0.09) / 3 = 0.1033$$

$$\text{Error} \approx 0.004$$

Tiny error is not an issue it is due to the noise and lower modulo. In real cases as modulo val increases this becomes negligible.

Real case

Full layer weights are transformed as polynomial

$$\{a, b, c\} \rightarrow (a + bx + cx^2) \times \Delta$$

secret key will be a Polynomial

$$s(x) \in R_q$$

$$R_q = \mathbb{Z}_q[x] / x^{N+1}$$

Public key

$a(x) \rightarrow$ random polynomial

evaluation key

basically error

$e(x) \rightarrow$ Poly with lower degree

$$b(x) = -a(x) \cdot s(x) + e(x)$$

$$\text{Pub key} = (a, b) = (a(x), b(x))$$

$$\text{P.T } m(x) = 288 + 325x + 471x^2$$

another random $u(x)$ and $e_1(x)$ $e_2(x)$

CT components

$$c_0 = b \cdot u + e_1 + m$$

$$c_1 = a \cdot u + e_2$$

$$CT_i = (c_0, c_1)$$

Decryption:

$$m'(x) = x_0 + x_1 \cdot s$$

and later divided by Δ